

S.NO	QUESTION	CO	BL	MARKS
1	Describe environmental pollution			
	Definition Environmental pollution is the undesirable change or contamination of the physical, chemical, and biological characteristics of the environment due to the introduction of harmful substances or energy. It adversely affects human beings, animals, plants, and natural resources. Major Types of Environmental Pollution 1. Air Pollution Contamination of the atmosphere by harmful gases, smoke, dust, and particulate matter. Sources: Vehicles, industries, thermal power plants, burning of fuels. Effects: Respiratory diseases, global warming, acid rain, and reduced air quality. 2. Water Pollution Contamination of rivers, lakes, groundwater, and oceans by harmful substances. Sources: Domestic sewage, industrial wastes, agricultural chemicals, and oil spills. Effects: Water-borne diseases, death of aquatic organisms, eutrophication, and unsafe drinking water. 3. Soil Pollution Degradation or contamination of soil due to harmful chemicals and wastes. Sources: Pesticides, fertilizers, industrial wastes, plastics, and solid wastes. Effects: Loss of soil fertility, contamination of food, and groundwater pollution. 4. Noise Pollution Unwanted or excessive sound that causes discomfort and health problems. Sources: Vehicles, industries, construction activities, loudspeakers, and aircraft. Effects: Hearing loss, stress, sleep disturbance, and increased blood pressure. 5. Thermal Pollution Increase in the temperature of natural water bodies due to the discharge of heated water from industries and power plants. Effects: Reduction in dissolved oxygen, disturbance of aquatic ecosystems, and death of aquatic organisms. 6. Nuclear Pollution Pollution caused by the release of radioactive substances into the environment. Sources: Nuclear power plants, radioactive wastes, and nuclear accidents. Effects: Genetic mutations, cancer, radiation sickness, and long-term environmental damage. General Causes • Rapid industrialization • Population growth and urbanization • Excessive use of fossil fuels • Improper disposal of wastes • Deforestation • Excessive use of fertilizers and pesticides • Vehicle emissions • Mining and construction activities Control Measures • Use renewable and clean energy sources. • Control industrial emissions and treat wastewater. • Promote public transport and electric vehicles. • Follow Reduce, Reuse and Recycle principles.			

	• Properly collect and dispose of solid and hazardous wastes. • Reduce excessive use of fertilizers and pesticides. • Develop green belts and promote afforestation. • Create environmental awareness among people. • Enforce environmental laws and standards.			
2	Write composition and properties of solid waste management	3	2	10
	Composition and Properties of Solid Waste Management 1. Introduction Solid waste is any unwanted or discarded solid material generated from human activities such as residential, commercial, industrial, agricultural, and institutional activities. Solid waste management (SWM) is the systematic collection, transportation, processing, recycling, treatment, and safe disposal of solid waste to protect public health and the environment. 2. Composition of Solid Waste The composition of solid waste varies depending on the source, location, season, lifestyle, and economic conditions. Major components 1. Organic matter ○ Food waste ○ Vegetable and fruit waste ○ Garden waste ○ Leaves and other biodegradable materials 2. Paper and cardboard ○ Newspapers ○ Office paper ○ Packaging materials ○ Cardboard boxes 3. Plastics ○ Plastic bags ○ Bottles ○ Packaging materials ○ Plastic containers 4. Glass ○ Bottles ○ Containers ○ Broken glass 5. Metals ○ Iron ○ Steel ○ Aluminium ○ Cans and metal containers 6. Textiles ○ Clothes ○ Cotton materials ○ Synthetic fabrics 7. Inert materials ○ Dust ○ Ash ○ Stones ○ Bricks ○ Construction debris 8. Other wastes ○ Rubber			

- Leather
- E-waste
- Household hazardous waste

Typical composition flow chart

Solid Waste



Organic Waste + Paper + Plastics + Glass + Metals + Textiles + Inert Materials + Others

3. Important Properties of Solid Waste

The properties of solid waste are mainly classified into:

A. Physical Properties

1. Density

Density is the mass of waste per unit volume.

Density = mass/volume

It is important for designing **collection vehicles, storage containers, and landfill facilities.**

2. Moisture Content

It is the amount of water present in solid waste.

High moisture content affects:

- Transportation
- Incineration
- Composting
- Calorific value

3. Particle Size and Size Distribution

The size of waste particles affects **processing, separation, composting, and recycling.**

4. Specific Weight

It is the weight of waste per unit volume.

5. Field Capacity

It is the total amount of moisture that a waste material can retain under gravitational conditions.

B. Chemical Properties

1. Proximate Analysis

It includes:

- Moisture
- Volatile matter
- Fixed carbon
- Ash

2. Ultimate Analysis

It determines the percentage of:

- Carbon
- Hydrogen
- Oxygen
- Nitrogen
- Sulphur
- Ash

3. Calorific Value

It is the amount of heat released when a unit quantity of waste is completely burned.

It is important in **incineration and waste-to-energy processes.**

4. pH

The pH indicates whether the waste is acidic or alkaline.

	5. Carbon-Nitrogen Ratio (C/N Ratio) It is important for composting and biological decomposition. C. Biological Properties Biological properties describe the ability of organic waste to undergo decomposition by microorganisms. Important characteristics include: • Biodegradability • Organic matter content • Microbial activity • Rate of decomposition • Odour generation Biodegradable wastes such as food, vegetables, leaves, and garden waste can be converted into compost. 4. Importance in Solid Waste Management Knowledge of the composition and properties of solid waste helps in: • Designing collection and transportation systems • Selecting suitable treatment methods • Recycling and resource recovery • Composting • Incineration • Designing sanitary landfills • Estimating energy recovery • Reducing environmental pollution			
3	Explain types of environmental pollution	3	2	10
	Introduction Environmental pollution is the contamination or undesirable change in the environment due to harmful substances or activities. It affects human health, plants, animals, and ecosystems. The major types of environmental pollution are: 1. Air Pollution Air pollution is the contamination of the atmosphere by harmful gases, smoke, dust, and particulate matter. Sources: • Vehicle emissions • Industries • Thermal power plants • Burning of coal, petrol, diesel and waste Effects: • Respiratory diseases • Global warming • Acid rain • Reduced visibility 2. Water Pollution Water pollution is the contamination of rivers, lakes, groundwater, and oceans by harmful substances. Sources: • Domestic sewage • Industrial wastewater • Agricultural fertilizers and pesticides • Oil spills			

	Effects: • Water-borne diseases • Death of aquatic organisms • Eutrophication • Unsafe drinking water 3. Soil Pollution Soil pollution is the contamination or degradation of soil due to harmful chemicals and wastes. Sources: • Excessive fertilizers and pesticides • Industrial wastes • Plastics • Solid waste • Mining activities Effects: • Loss of soil fertility • Reduced crop productivity • Contamination of food • Groundwater pollution 4. Noise Pollution Noise pollution is the presence of unwanted or excessive sound that causes discomfort and health problems. Sources: • Vehicles • Industries • Construction activities • Loudspeakers • Aircraft Effects: • Hearing loss • Stress and irritation • Sleep disturbance • High blood pressure 5. Thermal Pollution Thermal pollution is the undesirable increase in the temperature of water bodies due to the discharge of hot water. Sources: • Thermal power plants • Nuclear power plants • Industrial cooling water Effects: • Reduction in dissolved oxygen • Disturbance of aquatic ecosystems • Fish mortality • Changes in aquatic organisms 6. Nuclear Pollution Radioactive pollution is the contamination of the environment by harmful radioactive substances. Sources: • Nuclear power plants • Radioactive wastes			
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	• Nuclear accidents • Medical and research facilities Effects: • Cancer • Genetic mutations • Radiation sickness • Damage to living organisms			
4	Explain collecting and transporting process of solid waste management	3	2	10
	Introduction Collection and transportation are important steps in solid waste management. The collected waste should be safely transported from the source of generation to treatment, processing, or final disposal facilities. 1. Collection of Solid Waste Collection is the process of gathering solid waste from its sources and transferring it to a suitable collection point or vehicle. Sources of collection • Residential areas • Commercial establishments • Markets • Industries • Institutions • Hospitals • Streets and public places Methods of Collection 1. Door-to-Door Collection • Waste is collected directly from individual houses or buildings. • Separate containers can be used for wet and dry waste. • It helps in source segregation. 2. Community Bin Collection • Waste is deposited by people into common containers. • Collection vehicles periodically empty these containers. 3. Kerbside Collection • Residents place waste containers near the roadside. • Collection vehicles collect the waste at scheduled times. 4. Block Collection • A collection vehicle stops at a fixed location. • Residents bring their waste to the vehicle. 2. Transportation of Solid Waste Transportation is the movement of collected solid waste from collection points to transfer stations, processing plants, treatment facilities, or disposal sites. Types of Transportation 1. Direct Transportation • Waste is transported directly from the collection area to the disposal or treatment site. • Suitable when the disposal site is relatively close. 2. Transportation through Transfer Stations • Small collection vehicles bring waste to a transfer station. • Waste is transferred into larger vehicles. • Large vehicles transport it to the treatment or disposal facility. • This method is economical when disposal sites are far away.			

	Vehicles Used • Hand carts • Tricycles • Auto tippers • Tractor-trailers • Compactor trucks • Dumper trucks • Large haulage vehicles 3. Important Requirements An efficient collection and transportation system should: • Provide regular and timely collection. • Ensure proper segregation at source. • Prevent littering during transportation. • Use covered vehicles to prevent waste from falling out. • Control odour and dust. • Avoid leakage of liquid waste. • Select economical collection routes. • Maintain collection vehicles properly. • Provide suitable protective equipment for workers. Flow Chart Waste Generation ↓ Segregation at Source ↓ Primary Collection ↓ Collection Vehicle ↓ Transfer Station (if required) ↓ Transportation ↓ Treatment / Processing ↓ Final Disposal			
5	Describe Cause, effects and control measures of marine pollution and thermal pollution	3	2	10
	Marine Pollution and Thermal Pollution Causes, Effects and Control Measures – 10 Marks Answer A. Marine Pollution Definition Marine pollution is the introduction of harmful substances, wastes, or energy into seas and oceans, causing undesirable changes in marine water and affecting marine organisms and ecosystems. Causes of Marine Pollution 1. Domestic sewage – Untreated sewage discharged into seas. 2. Industrial wastes – Chemicals, heavy metals, and toxic substances enter marine water. 3. Oil spills – Leakage from ships, tankers, offshore drilling, and pipelines. 4. Plastic waste – Plastic bags, bottles, fishing nets, and other plastics accumulate in oceans. 5. Agricultural runoff – Fertilizers and pesticides are carried by rivers into the sea.			

	6. Mining activities – Mining wastes and heavy metals contaminate marine environments. 7. Ship activities – Waste oil, sewage, and garbage from ships pollute seawater. 8. Radioactive wastes – Improper disposal can introduce radioactive materials into marine environments. Effects of Marine Pollution • Death of fish and other aquatic organisms. • Reduction in marine biodiversity. • Eutrophication due to excess nutrients. • Oil films reduce oxygen exchange at the water surface. • Plastic ingestion and entanglement harm marine animals. • Bioaccumulation of toxic substances in marine food chains. • Damage to coral reefs and coastal ecosystems. • Health risks to humans consuming contaminated seafood. • Economic losses to fishing and tourism industries. Control Measures 1. Treat domestic sewage before discharge. 2. Treat industrial wastewater properly. 3. Prevent oil spills and improve ship safety. 4. Reduce the use of single-use plastics. 5. Promote recycling and proper waste disposal. 6. Control agricultural runoff by using fertilizers and pesticides carefully. 7. Enforce regulations on dumping wastes into oceans. 8. Conduct regular monitoring of marine water quality. 9. Promote public awareness about marine conservation. 10. Develop emergency plans for oil-spill control. B. Thermal Pollution Definition Thermal pollution is the undesirable change in the temperature of natural water bodies, mainly caused by the discharge of heated water from industries and power plants. Causes of Thermal Pollution 1. Thermal power plants – Hot cooling water is discharged into rivers and lakes. 2. Nuclear power plants – Large quantities of cooling water may be released at elevated temperatures. 3. Industrial processes – Industries such as steel, chemical, and petroleum industries discharge heated water. 4. Domestic sewage – Warm wastewater can increase water temperature. 5. Deforestation – Removal of vegetation along water bodies increases solar heating. 6. Dams and reservoirs – Changes in water circulation can alter natural temperature conditions. Effects of Thermal Pollution • Decrease in dissolved oxygen (DO) in water. • Death of fish and other aquatic organisms. • Disturbance of aquatic ecosystems. • Changes in the growth and reproduction of aquatic organisms. • Increase in the growth of some unwanted microorganisms and algae.			
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	- Thermal shock may occur when organisms experience sudden temperature changes. - Changes in species composition and biodiversity. Control Measures Cooling towers – Cool heated water before discharge. Cooling ponds – Store and cool wastewater naturally. Spray ponds – Reduce water temperature through evaporation. Recirculation of cooling water – Reuse water instead of continuously discharging it. Heat exchangers – Recover and reuse waste heat. - Maintain vegetation along water bodies to provide shade. - Follow proper standards for thermal discharge. - Regularly monitor the temperature and dissolved oxygen of receiving water bodies.			
6	Describe solid waste management	3	3	10
	Definition Solid Waste Management (SWM) is the systematic process of collection, segregation, storage, transportation, processing, recycling, treatment, and safe disposal of solid waste in order to protect human health and the environment. Sources of Solid Waste Residential waste – Food waste, paper, plastics, etc. Commercial waste – Waste from shops, markets, hotels, and offices. Industrial waste – Process wastes, ash, chemicals, etc. Institutional waste – Waste from schools, colleges, and government offices. Agricultural waste – Crop residues, animal waste, etc. Construction waste – Concrete, bricks, soil, wood, etc. Biomedical waste – Waste generated from hospitals and healthcare facilities. Steps in Solid Waste Management 1. Waste Generation Solid waste is generated from households, industries, commercial areas, institutions, and other activities. 2. Segregation at Source Waste should be separated at the point of generation into: Wet/biodegradable waste Dry/recyclable waste Domestic hazardous waste 3. Storage Waste is temporarily stored in suitable containers before collection. Containers should be properly covered and maintained. 4. Collection Waste is collected from houses, streets, markets, institutions, and other sources. Methods include: - Door-to-door collection - Community bin collection - Kerbside collection 5. Transportation Collected waste is transported using suitable vehicles to transfer stations, treatment facilities, or disposal sites.			

	6. Processing and Treatment a) Recycling: Materials such as paper, plastic, glass, and metals are recovered and reused. b) Composting: Biodegradable organic waste is converted into useful compost by microorganisms. c) Anaerobic digestion: Organic waste is decomposed in the absence of oxygen to produce biogas and digestate. d) Incineration: Combustible waste is burned at high temperature, reducing its volume and potentially recovering energy. 7. Final Disposal The remaining waste that cannot be recovered or treated is safely disposed of, mainly in a sanitary landfill. Flow Chart Waste Generation ↓ Segregation at Source ↓ Storage ↓ Collection ↓ Transportation ↓ Processing / Recycling / Composting / Energy Recovery ↓ Sanitary Landfill Objectives of Solid Waste Management • Protect public health. • Prevent air, water, and soil pollution. • Reduce the quantity of waste requiring disposal. • Recover useful materials and energy. • Maintain clean and healthy surroundings. • Promote the principles of Reduce, Reuse and Recycle (3Rs).			
7	Describe Cause, effects and control measures of air pollution and water pollution	3	2	10
	A. Air Pollution Definition Air pollution is the contamination of air by harmful gases, particulate matter, smoke, dust, and other substances in concentrations that can adversely affect humans, animals, plants, and the environment. Causes of Air Pollution 1. Vehicular emissions – Cars, buses, trucks, and two-wheelers release harmful gases. 2. Industrial emissions – Industries release smoke, dust, and gases such as SO₂ and NO_x. 3. Burning of fossil fuels – Coal, petrol, diesel, and natural gas produce pollutants. 4. Burning of waste – Burning plastics and other wastes releases toxic gases.	3	2	10

	5. Construction activities – Produce dust and particulate matter. 6. Agricultural activities – Burning of crop residues and use of certain chemicals contribute to air pollution. 7. Deforestation – Reduces the natural removal of carbon dioxide and particulate matter. 8. Natural sources – Forest fires, dust storms, and volcanic eruptions can also cause air pollution. Effects of Air Pollution On human health: • Respiratory diseases • Asthma and breathing difficulties • Eye and throat irritation • Cardiovascular problems On the environment: • Global warming and climate change • Acid rain • Damage to vegetation and crops • Reduced visibility • Damage to buildings and monuments Control Measures 1. Use public transport and carpooling. 2. Promote electric and low-emission vehicles. 3. Use cleaner fuels and renewable energy. 4. Install filters, scrubbers, and electrostatic precipitators in industries. 5. Control dust from construction sites. 6. Avoid open burning of waste. 7. Promote afforestation and green belts. 8. Regularly maintain vehicles and industrial equipment. 9. Enforce air-quality standards and emission controls. 10. Create public awareness about air pollution. B. Water Pollution Definition Water pollution is the contamination of surface water and groundwater by harmful physical, chemical, or biological substances, making water unsuitable for its intended uses. Causes of Water Pollution 1. Domestic sewage – Untreated sewage contains organic matter and pathogens. 2. Industrial wastewater – May contain chemicals, heavy metals, and toxic substances. 3. Agricultural runoff – Fertilizers and pesticides enter water bodies. 4. Oil spills – Oil from ships and offshore activities contaminates water. 5. Plastic and solid waste – Improper disposal causes contamination of rivers, lakes, and oceans. 6. Mining activities – Release sediments and harmful substances. 7. Thermal discharges – Hot water from power plants and industries changes water temperature. 8. Soil erosion – Increases sediment load in rivers and reservoirs.			
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	Effects of Water Pollution • Water-borne diseases such as cholera and diarrhoea. • Reduction in dissolved oxygen. • Death of fish and other aquatic organisms. • Eutrophication due to excess nutrients. • Unpleasant taste, colour, and odour. • Contamination of groundwater. • Bioaccumulation of toxic substances. • Loss of aquatic biodiversity. • Water becomes unsuitable for drinking, agriculture, and other uses. Control Measures 1. Treat domestic sewage before discharge. 2. Treat industrial wastewater using suitable treatment systems. 3. Reduce excessive use of fertilizers and pesticides. 4. Prevent dumping of plastics and solid waste into water bodies. 5. Control oil spills and provide proper spill-response systems. 6. Promote recycling and proper waste disposal. 7. Control soil erosion and agricultural runoff. 8. Reuse treated wastewater wherever appropriate. 9. Regularly monitor water quality. 10. Enforce pollution-control regulations and create public awareness.			
8	Describe Nuclear hazards in detail.	3	2	10
	1. Introduction Nuclear hazards are harmful effects on humans, animals, plants, and the environment caused by exposure to ionizing radiation or radioactive materials. Radioactive materials are used in nuclear power plants, medical facilities, industries, research laboratories, and agriculture. Improper handling, accidents, or disposal of radioactive materials can create serious hazards. 2. Sources of Nuclear Hazards 1. Nuclear Power Plants Nuclear reactors produce radioactive materials during the generation of electricity. Accidents or improper management can result in radiation release. 2. Nuclear Weapons Testing and use of nuclear weapons can release large quantities of radioactive materials into the atmosphere. 3. Radioactive Waste Waste from nuclear power plants, hospitals, laboratories, and industries may remain radioactive for long periods. 4. Medical Applications Radioactive substances are used in: • Cancer treatment • Diagnostic procedures • Medical research Improper handling can expose workers and patients to radiation. 5. Industrial Applications Radioisotopes are used in: • Industrial radiography • Thickness measurement • Leak detection			

- Material testing

6. Mining and Processing of Uranium

Mining, processing, and transportation of uranium-containing materials can expose workers and surrounding environments to radiation.

7. Nuclear Accidents

Accidents at nuclear facilities can release radioactive substances into air, water, and soil.

3. Types of Radiation

The important types of ionizing radiation are:

Alpha (α) Radiation

- Has low penetrating power.
- Can be dangerous when radioactive material enters the body through food, water, or inhalation.

Beta (β) Radiation

- Has greater penetrating power than alpha radiation.
- Can affect the skin and internal organs.

Gamma (γ) Radiation

- Has very high penetrating power.
- Can travel long distances through materials.
- Requires suitable shielding such as concrete or lead.

Neutron Radiation

- Produced in nuclear reactions.
- Has significant penetrating power and can be highly hazardous.

4. Effects of Nuclear Hazards

A. Effects on Human Health

Short-Term Effects

High exposure to radiation can cause:

- Skin burns
- Nausea and vomiting
- Fatigue
- Radiation sickness
- Damage to body tissues

Long-Term Effects

Long-term or repeated exposure may increase the risk of:

- Cancer
- Genetic and reproductive effects
- Cataracts
- Organ and tissue damage

The severity of effects depends on **radiation type, dose, duration of exposure, and sensitivity of the affected tissue.**

B. Effects on the Environment

1. **Soil contamination** – Radioactive substances can remain in soil for long periods.
2. **Water contamination** – Radioactive materials can enter rivers, lakes, and groundwater.
3. **Air contamination** – Radioactive particles and gases can spread through the atmosphere.
4. **Food-chain contamination** – Radioactive substances can accumulate in plants and animals.

	5. Damage to ecosystems – Radiation can affect plants, animals, and microorganisms. 6. Long-term environmental persistence – Some radionuclides have long half-lives and may remain hazardous for extended periods. 5. Control and Preventive Measures 1. Proper Shielding Use suitable shielding materials such as: • Lead • Concrete • Water 2. Reduce Exposure Time Workers should spend the minimum necessary time near radiation sources. 3. Increase Distance Increasing the distance from a radiation source generally reduces radiation exposure. 4. Safe Handling and Storage Radioactive materials should be properly: • Stored • Labelled • Transported • Monitored 5. Proper Disposal of Radioactive Waste Radioactive waste must be treated and disposed of using approved methods to prevent environmental contamination. 6. Radiation Monitoring Radiation levels should be continuously or periodically monitored using suitable instruments. 7. Personal Protective Equipment Workers should use appropriate protective equipment and personal radiation-monitoring devices such as dosimeters. 8. Emergency Preparedness Nuclear facilities should have emergency plans for: • Radiation leaks • Fire • Equipment failure • Nuclear accidents 9. Regular Inspection and Maintenance Nuclear facilities and equipment should be regularly inspected and maintained according to safety requirements. 10. Public Awareness and Training Workers and nearby communities should be trained about radiation risks and emergency procedures. 6. Basic Principle of Radiation Protection The three important principles are: TIME → Minimize time of exposure DISTANCE → Maximize distance from the source SHIELDING → Use appropriate shielding			
9	Describe Cause, effects and control measures of Noise pollution and soil pollution	3	2	10

	A. Noise Pollution Definition Noise pollution is the presence of unwanted, excessive, or unpleasant sound in the environment that causes discomfort and adverse effects on humans and other living organisms. Causes of Noise Pollution 1. Road traffic – Horns, engines, and heavy vehicles produce noise. 2. Industries – Heavy machinery, generators, compressors, and turbines generate noise. 3. Construction activities – Drilling, blasting, and construction equipment produce high noise levels. 4. Aircraft – Take-off and landing of aircraft create high-intensity noise. 5. Railways – Trains, horns, and railway operations generate noise. 6. Loudspeakers – Used during functions, festivals, and public events. 7. Household appliances – Mixers, generators, televisions, and other equipment can contribute to noise. 8. Firecrackers – Produce sudden and high-intensity noise. Effects of Noise Pollution On human health: • Hearing impairment or hearing loss • Headache and irritation • Stress and anxiety • Sleep disturbance • Increased blood pressure • Reduced concentration and productivity On animals and environment: • Disturbs animal communication • Affects breeding and feeding patterns • Causes animals to move away from their habitats Control Measures 1. Maintain and service vehicles and machinery regularly. 2. Use silencers and noise-control devices. 3. Develop green belts around industries and roads. 4. Construct sound barriers near highways and sensitive areas. 5. Restrict unnecessary use of horns. 6. Control loudspeakers and firecrackers. 7. Use low-noise machinery and equipment. 8. Properly plan industrial and residential areas. 9. Provide hearing protection to workers exposed to high noise. 10. Enforce prescribed noise standards and create public awareness. B. Soil Pollution Definition Soil pollution is the contamination or degradation of soil due to the addition of harmful chemicals, wastes, and other pollutants, resulting in a reduction in soil quality and fertility. Causes of Soil Pollution 1. Excessive use of fertilizers – Can cause chemical accumulation and soil degradation.			
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	Pesticides and herbicides – May leave harmful residues in soil. Industrial wastes – Chemicals and heavy metals may contaminate soil. Improper disposal of solid waste – Plastics, metals, and other wastes accumulate in soil. Mining activities – Mining generates waste and can introduce toxic substances. Sewage and sludge – Improperly treated sewage may contain harmful substances. Oil and petroleum products – Leakage and spills contaminate soil. E-waste – Improper disposal can release metals and other hazardous substances. Effects of Soil Pollution - Reduction in soil fertility. - Reduced crop productivity. - Destruction of beneficial soil microorganisms. - Contamination of agricultural products. - Groundwater contamination. - Bioaccumulation of toxic substances. - Harm to plants, animals, and humans. - Soil degradation and loss of soil quality. Control Measures - Use fertilizers and pesticides in recommended quantities. - Promote organic farming and biological pest control. - Properly collect, treat, and dispose of industrial wastes. - Follow the 3Rs – Reduce, Reuse and Recycle. - Avoid open dumping of solid waste. - Properly manage hazardous and e-waste. - Treat sewage before using or disposing of it. - Control mining activities and restore mined land. - Prevent oil and chemical leakage. - Promote afforestation and soil conservation.			
10	Explain role of individual in prevention of pollution	3	2	10
	Introduction Pollution prevention is not only the responsibility of governments and industries. Every individual has an important role in reducing pollution and protecting the environment. Small changes in daily activities can collectively produce a significant positive effect. Role of Individuals in Prevention of Pollution Reduce, Reuse and Recycle - Follow the 3Rs – Reduce, Reuse and Recycle. - Avoid unnecessary consumption. - Reuse bags, bottles, containers, and other useful materials. - Separate recyclable materials from other waste. Proper Waste Disposal - Do not throw waste on roads, streets, or into water bodies. - Dispose of waste in appropriate bins. - Segregate wet waste, dry waste, and hazardous waste at source. Reduce Plastic Usage - Avoid single-use plastic bags, cups, and bottles. - Use reusable cloth or durable bags.			

	• Participate in plastic collection and recycling programs. 4. Use Public Transport • Prefer buses, trains, cycling, or walking whenever possible. • Carpool with others. • Maintain vehicles properly to reduce harmful emissions. 5. Save Energy • Switch off lights, fans, and electrical appliances when not required. • Use energy-efficient appliances. • Prefer renewable energy such as solar energy where feasible. 6. Prevent Water Pollution • Do not throw plastics, chemicals, oils, or other wastes into drains and water bodies. • Use water carefully. • Avoid excessive use of detergents and chemical products. 7. Prevent Soil Pollution • Use fertilizers and pesticides responsibly. • Avoid dumping chemicals, batteries, paints, and e-waste on the soil. • Dispose of hazardous waste through appropriate collection systems. 8. Control Noise Pollution • Avoid unnecessary use of vehicle horns. • Keep the volume of loudspeakers and music at reasonable levels. • Avoid unnecessary firecrackers. • Maintain vehicle silencers properly. 9. Plant and Protect Trees • Plant trees and maintain green areas. • Avoid unnecessary cutting of trees. • Participate in tree-plantation programs. 10. Create Environmental Awareness • Educate family members, friends, and the community about pollution prevention. • Participate in environmental campaigns. • Encourage others to adopt environmentally responsible practices. 11. Avoid Open Burning • Do not burn plastics, leaves, or household waste in the open. • Use proper waste collection and composting systems. 12. Follow Environmental Rules • Follow local waste-management and pollution-control regulations. • Report illegal dumping and serious pollution incidents to the appropriate authorities.			
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